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Three-year dynamics of methylmercury production in Hg^{2+}-spiked paddy soils: Mercury speciation, microbial communities, and rice contamination

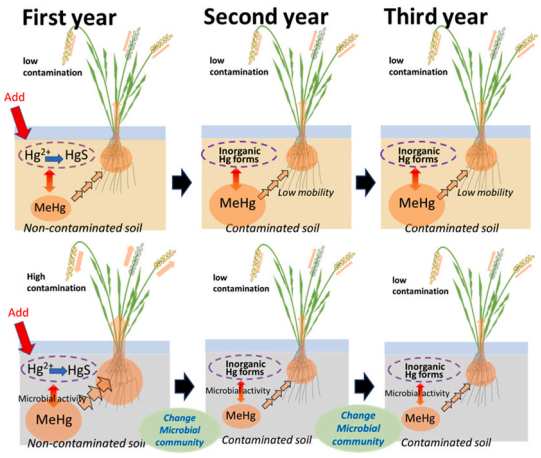
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HIGHLIGHTS

- Hg forms and microbial communities in soil after 10 ppm Hg^{2+} input were studied.
- Rice grain Hg level was affected by both MeHg concentration and Hg mobility in soil.
- Results from sequential extraction could not explain MeHg formation in soil.
- The MeHg/T-Hg ratio in soil correlated positively with *hgcAB* gene abundance.

GRAPHICAL ABSTRACT



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ABSTRACT

We investigated three-year changes in soil mercury (Hg) pools, methylmercury (MeHg) production, rice contamination, and microbial communities after a single Hg^{2+} addition to two soils (Soil I and Soil II). In Soil I, total Hg (T-Hg) concentration of brown rice grain was 0.150 ± 0.023 mg/kg (n = 143) in 2015 and increased to 0.233 ± 0.080 (n = 135) and 0.240 ± 0.118 mg/kg (n = 225) in 2016 and 2017. In Soil II, T-Hg declined from 0.530 ± 0.101 (n = 130) in 2015 to 0.124 ± 0.059 (n = 213) and 0.168 ± 0.059 mg/kg (n = 200) in 2016 and 2017. Variations in T-Hg concentrations in rice grains cultivated in the two soils showed a relationship with soil MeHg concentrations within the same soil, but not between different soils. Sequential extraction, which partitioned soil Hg into seven fractions, indicated that Soil II contained a higher proportion of water-extractable Hg. This finding suggests that the mobility of Hg may have influenced the level of Hg contamination in rice grains. The proportion of Hg sulfide peaked approximately one month after the addition of Hg^{2+} in both soils, then

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decreased over time. In contrast, the fractions of organic-bound and elemental Hg tended to increase over time. In soil II, where DNA extraction was successful, microbial communities showed no clear differences at the phylum level between the Hg-added and non-added samples, but distinct shifts were observed at lower taxonomic levels. Metagenomics showed that the MeHg/T-Hg ratio correlated positively with *hgcAB* gene abundance ($r = 0.85$, $P < 0.05$), while *merA/merB* showed no clear relationship.

1. Introduction

Recently, rice consumption has been linked to mercury (Hg) exposure in humans (M. Liu et al., 2019). Hg contamination in rice occurs when inorganic Hg in paddy soils is converted to monomethylmercury (MeHg), which rice plants then absorb. Feng et al. have extensively studied MeHg contamination in rice and its mitigation. Their findings were recently compiled and published as a comprehensive book (Feng et al., 2024). Studies on Hg contamination in rice initially focused on polluted areas in China's Guizhou Province, which has a history of mining Hg as cinnabar (α-HgS) (Feng and Qiu, 2008). However, new Hg contamination is emerging as a result of human activities such as artisanal and small-scale gold mining (ASGM) (Esdaile and Chalker, 2018; Obrist et al., 2018). Hg contamination of the environment, including paddy fields, as a result of ASGM activity has become a major concern, particularly in Southeast Asia, which is known for rice cultivation (Novirsa et al., 2019, 2020; Soe et al., 2022). Hg pollution from ASGM arises from various sources, including Hg^{2+} generated during the mixing of gold ore, water, and Hg, as well as Hg^0 released to the atmosphere during the heating of amalgamated gold. These Hg species affect the atmosphere and aquatic environments, eventually accumulating in plants and soil via air and water pathways (Novirsa et al., 2019; Tomiyasu et al., 2013, 2019, 2020). Consequently, regions that were previously unaffected by Hg contamination may now face severe Hg pollution. Our earlier work found that paddy soils in an ASGM area in Indonesia had a wide range of total Hg (T-Hg) concentrations ranging from 0.38 to 57.4 mg/kg (Tomiyasu et al., 2013). Because soil remains in place for a long time, it is necessary to examine how Hg-contaminated soil affects the environment over time, from the moment of contamination.

One way to reduce Hg contamination in rice grains is to understand and control MeHg formation in the soil. MeHg synthesis in flooded paddy soil is widely recognized to be caused by anaerobic bacteria such as sulfate-reducing bacteria (SRB) (Barkay et al., 2003). Recent studies have confirmed that *Geobacter* and *Anaerolinea* are also major Hg-methylating bacteria, and it has been reported that their interactions, rather than individual activity, are particularly effective (Pu et al., 2024; Zhong et al., 2024). Metagenome analysis was used to identify the microbes containing the *hgcAB* genes, which drive MeHg production (Parks et al., 2013). Recent studies have also documented the use of soil amendments such as biochar to reduce Hg contamination in rice (Li et al., 2022; Yang et al., 2021; Zheng et al., 2024). However, it is unclear how the use of these soil amendments, as well as new Hg contamination, could modify the growth environment for bacteria in the soil. The difference in behavior between newly deposited Hg and existing Hg also has long been a subject of interest (Liu et al., 2022b). Currently, we lack sufficient information to assess the long-term effects of new Hg contamination in soil or the application of soil amendments to Hg-contaminated soil on bacterial communities and MeHg production.

In addition, the difficulty in assessing Hg contamination in soil lies in the forms of inorganic Hg present. To date, many studies have evaluated soil Hg concentrations using T-Hg and MeHg concentrations. Only approximately 0.1 % of the T-Hg in soil exists as MeHg. However, the amount of MeHg produced by microbes in soils and sediments is determined by the amount of inorganic Hg that is efficiently convertible by these microbes (J. Liu et al., 2019). Therefore, procedures such as sequential extraction have been used to determine the type of inorganic Hg present (Ali et al., 2024; O'Connor et al., 2019; Reis et al., 2016). Chemical species such as dissolved organic matter and sulfide ions in soil have been examined for their effects on inorganic Hg solubilization and microbial methylation processes (Graham et al., 2012; Li et al., 2017, 2019; Liu et al., 2022a, 2023). Recently, bioavailable Hg has been investigated using diffusion membranes (Huang et al., 2020; Ndu et al., 2018). However, understanding the behavior of Hg in soils with complex compositions remains difficult.

We previously investigated Hg contamination in rice by introducing Hg^{2+} into uncontaminated soil and growing rice in pots with the Hg-contaminated soil (Kodamatani et al., 2020). In this study, therefore, we continued to use this Hg^{2+}-added soil in a 3-year rice-growing experiment to investigate not only rice contamination but also MeHg formation behavior in the soil, changes in microbial communities, occurrence frequency of functional genes, and conversion of added Hg^{2+} to other forms. The findings of this study shed light on how new Hg contamination occurs and how it affects continuously cultivated rice paddies.

2. Materials and methods

2.1. Rice cultivation and sampling

In 2015, we evaluated MeHg production in soil and rice contamination across two soil type Soil I and Soil II. Soil I was a commercial soil formed artificially with ~5 mm granularity for plant nurseries, and Soil II was collected from a paddy field at the Kagoshima University Experimental Farm. The soil composition is shown in Table S1. Twenty kilograms of each soil were placed in pots. To produce a concentration of 10 mg/kg of dry weight, water was added, and the soil was thoroughly mixed before adding Hg^{2+} (as $HgCl_2$). Experimental systems designed to simulate new Hg contamination in soils should determine the concentration and chemical form of Hg to be added based on insights from Hg mass balance studies and potential sources such as localized pollution from industrial effluents (Wadle et al., 2024; Wang et al., 2019; Xia et al., 2021). In the present study, the Hg addition concentration was set to represent a highly contaminated environment, reflecting field conditions in ASGM area of Indonesia, where wastewater from gold extraction using gold ore and metallic Hg is directly discharged into paddy fields, and most of the Hg in the wastewater exists as Hg^{2+} (Tomiyasu et al., 2013, 2019). We placed these pots in three locations and cultivated rice plants in three locations: Greenhouse 1 (G1), Greenhouse 2 (G2), and the controlled-temperature room. The cultivation summary and results for 2015 have already been reported (Kodamatani et al., 2020). Part of those samples were used in this study.

To prepare for 2016 cultivation, the soil from 2015 was naturally dried after the rice harvest and sieved through a 2-mm sieve to remove roots. Similarly, soil after 2016 cultivation was used in 2017. Three seedlings (Taichung 65) were planted on the day after the water was added as in 2015. No further $HgCl_2$ was added in 2016 or 2017. During rice growth, waterlogging was maintained with well water, and fertilizers (sodium phosphate, ammonium sulfate, and potassium chloride) were applied according to the rice-growing conditions. In 2016 and 2017, one pot each of Hg^{2+}-supplemented soils I and II was prepared and cultivated solely at the G1 site. Throughout the three years, waterlogging began in June, rice was planted, panicles emerged at the end of August, and harvesting occurred in November. The three-year cultivation summary is shown in Table S2.

To prevent disruption of the soil environment caused by soil

sampling and to maintain the soil environment, we collected several grams of topsoil (~3 cm thick) with a plastic spoon once a month throughout the rice-growing season (June to November) for 3 years. Additionally, core samples were collected in July 2015 and 2017 to confirm the vertical distribution of Hg concentrations in the pot; data from 2015 was reported previously (Kodamatani et al., 2020).

Soil and plant samples were immediately freeze-dried to maintain the MeHg concentration in the soil (Kodamatani et al., 2017b). The dried soil samples were sieved through a 2-mm sieve and finely ground using an agate planetary ball mill (PM200, Verder Scientific, Germany) or Multi-bead shocker (MB3000, Yasui Kikai, Osaka-city, Japan). After drying, the samples were placed in double-layered plastic bags with zippers and stored at room temperature. To determine the T-Hg concentration in brown rice samples, each grain was removed from its hull, weighed, and used as a sample for measurement.

2.2. Hg measurements

T-Hg concentrations in the soil and rice plant samples were determined using a fully automated thermal decomposition cold-vapor atomic absorption spectroscopy system (MA3000, Nippon Instruments Co., Kyoto, Japan). The detection limit (MDL, n = 7, 3σ) for T-Hg was 7 pg. MeHg concentrations in soil samples were measured using a high-performance liquid chromatography-chemiluminescence detection technique (Kodamatani et al., 2012, 2017a; Kodamatani and Tomiyasu, 2013). The MDL of MeHg was 0.07 ng g^{-1} for 1 g of sample.

For MeHg determination in soils, 24 samples were analyzed in a single batch, including a method blank, calibration solutions, and duplicate analyses of the certified reference material (CRM) ERM CC580 (estuarine sediment; T-Hg, 132 ± 3 mg/kg; MeHg, 75 ± 4 μg/kg) for quality assurance and control. For the CRMs used in measuring T-Hg concentrations in soil and rice grain, ERM CC580 and NMIJ CRM 7402-a (Cod fish; T-Hg, 0.61 ± 0.02 mg/kg; MeHg, 0.58 ± 0.02 mg/kg) were employed respectively. The CRM values, which were measured in the same batch and fell within the certified range, confirmed the dependability of the measured T-Hg and MeHg values.

2.3. Soil properties

The soil properties were characterized as follows: The main mineral compositions were determined using semi-quantitative analytical X-ray fluorescence (ZSX-mini II, Rigaku Co., Tokyo, Japan), which yielded a total detected mineral composition of 100 %. Total carbon (T-C), total nitrogen (T-N), and total sulfur (T-S) were determined using organic elemental analysis (Flash 2000; Thermo Fisher Scientific, Waltham, MA, USA).

To analyze the eluted chemical species from the soil, 1 g of soil was shaken with 2.5 mL of ultrapure water for 30 min and centrifuged. The pH of the extracted solutions was measured using a pH meter (Horiba, Kyoto, Japan). The supernatant was filtered through a 0.45-μm hydrophilic PTFE filter and used as the sample solution. Dissolved organic carbon (DOC) and dissolved nitrogen concentrations in the extract were determined as non-purgeable organic carbon and total nitrogen, respectively, using a TOC-L analyzer (Shimadzu Corporation, Kyoto, Japan). The major anions and cations in the extracted solution were determined using an ion chromatography system (Dionex Integrion HPIC, Thermo Fisher Scientific).

2.4. Sequential extraction method for Hg fractionation analysis

We employed a seven-step fractionation extraction method, combining the three previously reported fractionation extraction methods (Fernández-Martínez et al., 2014; Kanjo et al., 2008; Pelcová et al., 2020; Zvěřina et al., 2014). Using six extraction solutions, soil Hg was sequentially extracted into seven fractions: step 1, water-extractable Hg; step 2, acid-soluble Hg; step 3, reducible Hg; step 4, organic-bound Hg; step 5, elemental Hg; step 6, Hg sulfide; and step 7, residual Hg. Details are shown in Text S1 and Table S3.

2.5. Microbial community and metagenomic analysis

Paddy soil DNA was extracted from 2.3 to 2.5 g freeze-dried soil using a DNeasy Power Max Soil Kit (QIAGEN, Hilden, Germany) as 5 mL solution following the manufacturer's protocol. The extracted DNA was used to perform 16S rRNA amplicon and shotgun metagenomic sequencing. These details are shown in Text S2.

2.6. Data availability

The 16S rRNA amplicon sequencing data were deposited in the DDBJ with accession codes DRR687441-DRR687454. Metagenome sequencing data were deposited in DDBJ with the accession codes DRR687455-DRR687463.

3. Results and discussion

3.1. Changes in T-Hg concentration in brown rice from 2015 to 2017

Rice was cultivated annually until 2017 using two types of soil to which Hg^{2+} was added in 2015. Brown rice grains cultivated in Soil I had a T-Hg concentration of 0.150 ± 0.023 mg/kg (n = 143) in 2015 and remained stable at 0.233 ± 0.080 mg/kg (n = 135) and 0.240 ± 0.118 mg/kg (n = 225) in 2016 and 2017, respectively. However, in Soil II, the concentration decreased substantially from a peak of 0.530 ± 0.101 mg/kg (n = 130) in 2015 to 0.124 ± 0.059 mg/kg (n = 213) and 0.168 ± 0.059 mg/kg (n = 200) in 2016 and 2017, respectively. The weight and T-Hg content of a single grain of harvested brown rice are shown in Fig. 1. The weight of rice grains varied over the years. This could be related to variations in environmental conditions, such as soil nutrients and temperature, which cause variations in growth. Our previous study found that although the concentration of Hg in brown rice grains grown in the same pot can vary several-fold depending on the panicle, there is a positive correlation between the weight of brown rice grains and the amount of Hg contained, implying that the concentration remains nearly constant (Kodamatani et al., 2020). In this experiment as well, a linear relationship was confirmed between the weight of rice grains and the amount of T-Hg. Therefore, we believe that poor rice grain growth is not the cause of annual variations in T-Hg concentrations of rice grains. Our previous study has confirmed that Hg in brown rice is predominantly MeHg (Kodamatani et al., 2020), suggesting that changes in brown rice concentrations reflect the concentration of MeHg in the soil.

3.2. Changes in MeHg concentration in the soils from 2015 to 2017

Fig. 2 shows the monthly changes in the MeHg/T-Hg ratio in each soil from 2015 to 2017 (T-Hg and MeHg concentrations are shown in Fig. S1, and the 2015 soil Hg data are identical to those reported in a previous paper (Kodamatani et al., 2020)). Previously, we observed that the vertical distribution of the T-Hg concentration in soil treated with Hg^{2+} in 2015 showed higher concentrations in the surface layers and lower concentrations in deeper layers. In 2017, the third year following Hg^{2+} addition, the vertical distribution showed approximately uniform T-Hg concentration from the upper to the lower layers (Fig. S2). Because the T-Hg concentration in the sampled surface layers fluctuated widely from year to year, we conducted the evaluation using the MeHg/T-Hg ratio.

In Soil I, which was artificially granulated to approximately 5 mm, we previously reported that MeHg detection began four months after flooding in 2015 (Kodamatani et al., 2020). However, beginning in the second year, the soil granules disintegrated, potentially promoting the development of a reductive environment, and MeHg production was

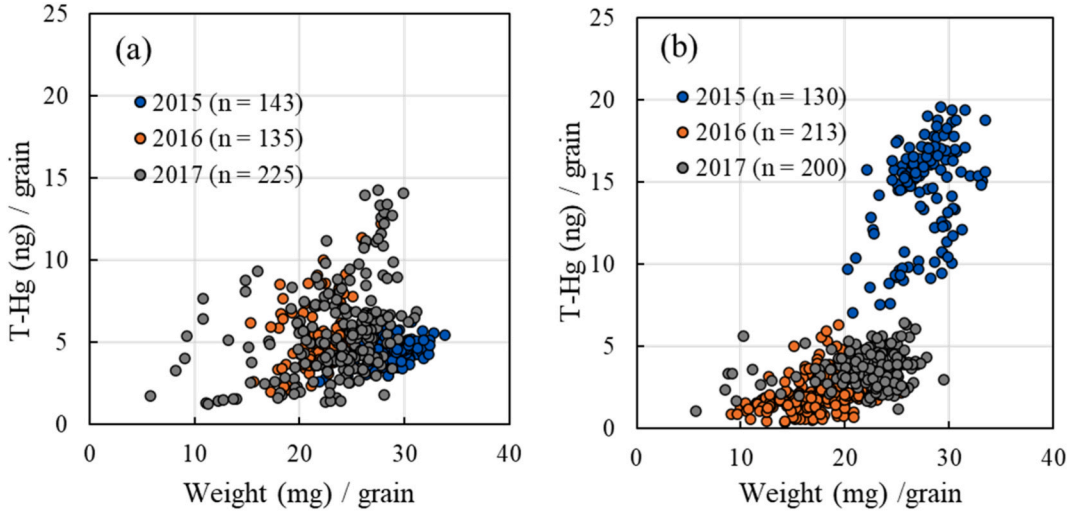


Fig. 1. Changes in weight and total Hg per grain of brown rice grown in Soil I (a) and Soil II (b) during three seasons from 2015 to 2017. The 2015 brown rice samples were collected from the G1.

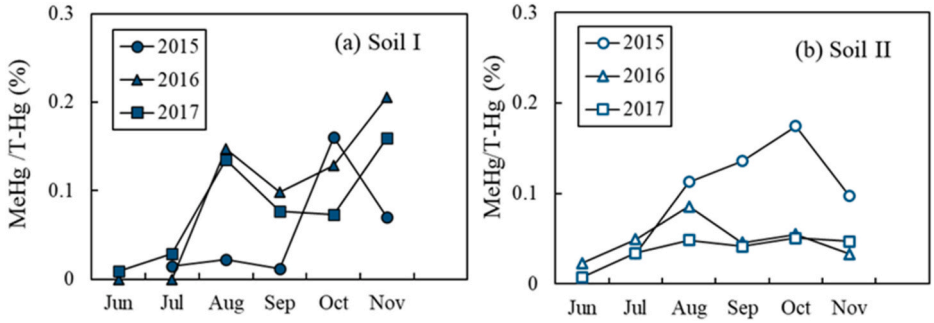


Fig. 2. Changes in the MeHg/T-Hg ratio in surface soils of (a) Soil I and (b) Soil II across three seasons, 2015 to 2017. Data for 2015 collected from the G1.

detected within one month of flooding. The MeHg/T-Hg ratio remained high during the growing seasons in 2016 and 2017 compared to that in 2015. In Soil II, the MeHg/T-Hg ratio peaked at 0.2 % in October 2015; however, it declined in 2016 and 2017 compared to that in 2015. Furthermore, in both soils, the MeHg/T-Hg ratio decreased each year before the onset of flooding. We consider that this decrease occurred because we allowed the soils to dry naturally between the end of the rice-growing season and the start of cultivation the following year, during which time MeHg likely decomposed (Kodamatani et al., 2017b).

Regarding the relationship between rice and soil MeHg/T-Hg ratio, given that the rice panicle usually emerges around the end of August and matures in about two months, the relationship between the MeHg/T-Hg ratio in each soil for each year (Fig. 2) and Hg concentration in harvested brown rice (Fig. 1) was consistent. When comparing the two soils, the proportion of soil MeHg in Soil I (2016 and 2017) was similar to Soil II (2015), implying similar levels of Hg in rice grain; however, the concentration of rice grain in Soil I was almost half of that of Soil II. Research is progressing on how carbon, nitrogen, and sulfur in soils affect MeHg in soils. Organic carbon functions as a nutrient source for microorganisms, and nitrogen and sulfur, depending on their chemical forms, are involved in microbial growth and respiration (Chen et al., 2023; Li et al., 2022, 2023; Liu et al., 2023). Furthermore, sulfide ions generated from sulfate ions used in respiration also affect the mobility of Hg (Li et al., 2017, 2022). Therefore, we examined the chemical compositions of the soil and its extracted solution (Table S4). In a previous study, we observed that the variations in the vertical distribution of Hg between Soils I and II might be attributed to differences in the soil organic matter concentrations. In addition, this experiment examined DOC, which was found to be higher in Soil II than in Soil I. The higher vertical distribution of T-Hg in the 2015 soil's surface layer and the lower distribution in the lower layer could be attributed to the high suspended organic matter and slow settling of Hg in the form of bound organic matter and Hg^{2+}. In contrast, concentrations in the upper and lower layers of soil core samples from the third year (2017) were almost constant (Fig. S2), despite T-C and DOC showing little change from 2015. This suggests that the form of Hg^{2+} has changed since 2015. In this experiment, granular chemical fertilizer (sodium phosphate, ammonium sulfate, and potassium chloride) was applied with the primary objective of promoting healthy rice growth. In the November 2016 samples, a clear increase in NH_4^+, SO_4^{2-}, and K^+ concentrations due to fertilization was confirmed (Table S4). However, it was not possible to examine temporal variations in ion concentrations or assess the impact of fertilizer containing SO_4^{2-} on MeHg production.

3.3. *Changes in Hg forms in soil*

The formation of MeHg from inorganic Hg is largely reliant on the forms of inorganic Hg (J. Liu et al., 2019). As shown in Fig. 2, the changes in MeHg formation behavior differed between Soils I and II. We hypothesized that the decrease in MeHg production in Soil II after the second year was due to the added Hg^{2+} acquiring a stable form, such as crystalline HgS, which is less likely to be utilized by microorganisms. Hg^{2+} introduced to paddy soil is expected to be bound to sulfide ions from reducing environments and organic matter. Furthermore, it would be interesting to investigate how the types of Hg formed in the soil change over time under both flooded and non-flooded conditions, which

represent the cycle of paddy fields. As described in Section 3.2, the vertical distribution of the T-Hg concentration changed, suggesting that the Hg state was modified. Therefore, sequential extraction method was employed to separately quantify the various forms of Hg present in the soil samples.

Table 1 shows the results of the fractionation analysis of soil samples. The total amount of eluted Hg in all steps relative to the total amount of Hg in soil could be assessed with recovery ratios of 83–112 % (Table S5). When Soil I and II are compared, it is clear that Soil II contains a higher percentage of Hg in the water-extractable and organic-bonded forms (Steps 1 and 4, respectively). These findings are consistent with the expected results given the higher concentrations of T-C and DOC in Soil II compared to Soil I (Table S4). As shown in Fig. 2, even though the percentage of MeHg produced in Soil I in 2016 and 2017 was similar to the percentage of MeHg produced in Soil II in 2015, the Hg concentration in rice grown in soil II in 2015 was approximately two times higher (Fig. 1), which can be attributed to the greater mobility of Hg species in the soil in pots under waterlogged conditions. Initially, we predicted that the decrease in the MeHg/T-Hg ratio in Soil II was the result of the added Hg^{2+} converting to stable chemical forms, such as HgS, over time. However, both soils had a high percentage of sulfide form (Step 6) in 2015, with the percentage of sulfide form (Step 6) decreasing and the percentage of organic-bonded and elemental forms (Steps 4 and 5) increasing with time. In addition to the soil's heterogeneity, the vertical distribution of T-Hg concentrations was changing. Therefore, changes in the fraction of Hg forms over time cannot be ignored. However, these results may reflect the oxidative degradation of HgS caused by drying after rice cultivation, the conversion to Hg^0 by microbial activity, and an increase in the proportion of organic-bound Hg due to rice roots remaining in the soil. In any event, the understanding of Hg forms obtained from the fractionation extraction method was insufficient in this experiment to comprehend the formation behavior of MeHg. Recently, a study was reported in which five mercury species with different isotopic masses (Hg^{2+}, Hg–DOM, nano-HgS, and Hg–FeS) were added to surface water, followed by sequential extraction of Hg accumulated in bottom sediments (Wadle et al., 2024). However, more than 90 % of the extracted Hg was detected as organic-bound Hg. The authors assumed that particulate and stable species such as HgS and Hg–FeS would be extracted in their original forms without transformation. This suggests that the results obtained from sequential extraction may not accurately reflect the actual forms of Hg present in the environment. However, it is interesting to note that about one month after the addition of Hg^{2+}, a high percentage of Hg in the soil converts to HgS. Classification by sequential extraction is incomplete and likely includes not only HgS but also other Hg forms as well; however, it is certain that these forms do not dissolve easily. The forms of Hg in soil, as well as the variations in its morphology caused by environmental changes, plus the assessment of its bioavailability, are very complex, and understanding them remains a challenge today. The presence of rice also made this experiment more challenging. Future experiments will focus solely on long-term changes in the soil.

3.4. Microbial communities in the soil

Several bacteria, including *Thermodesulfobacteriota* (known as SRBs) and *Bacillota,* are known to be the key players in the formation of MeHg from inorganic Hg in natural anaerobic soils and sediments (Barkay et al., 2003). Therefore, the microbial composition of soil is an important indicator for evaluating MeHg production. Moreover, the multi-year succession of the microbial community when previously non-Hg-contaminated soil experiences new Hg contamination is of great interest. The DNA was extracted from each soil sample (Table S6). Except for one sample, DNA concentrations from Soil I samples were below the detection limit (0.1 ng/μL). In contrast, the DNA concentrations in Soil II samples ranged from 0.9 to 3.3 ng/μL. Soil I had a high MeHg/T-Hg ratio from the second year onward, necessitating DNA extraction to better understand bacterial community changes. However, it is well known that the adsorption of DNA onto soil minerals can hinder extraction (Foysal and Salgar-Chaparro, 2024; Roose-Amsaleg et al., 2001). The study did not investigate whether the low DNA yield was due to a low-abundant microorganisms or soil composition interfering with extraction. In this study, only Soil II samples were used for further DNA analysis.

Microbial community compositions in Soil II samples were investigated using the 16S rRNA gene amplicon (Fig. 3). The phylum *Bacillota* dominated the microbial community in all 14 samples, accounting for approximately 60 %–80 % of the total (Fig. 3a). In the first year (2015), there were no clear differences in the phyla between the Hg-added and non-added samples. *Acidobacteria*, *Actinomycetota,* and *Chloroflexota* increased over two and three years. To quantify variations in the microbial community composition at lower taxonomic levels, we used non-metric multidimensional scaling (NMDS) analysis with amplicon sequence variant (ASV) proportions (Fig. 3b). Focusing on the first year (2015), a comparison of samples from the same months indicated distinct differences with and without Hg addition (H and C) (II.1.2015.07H and 07C, 09H and 09C, and 11H and 11C). In contrast, the influence of differences between growth environments G1 and G2 showed close plot locations (II.1.2015.09H and II.2.2015.09H). Over the course of the cultivation period, samples collected from the first and subsequent months (e.g., II.1.2015.07H–09H and 11H) exhibited a

Table 1
Results of Hg fractionation analysis of soil samples.

Sample ID[a]	Hg (%)[b]						
	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Step 7
	Water-extractable Hg	Acid-soluble Hg	Reducible Hg	Organic bound Hg	Elemental Hg	Hg sulfide	Residual Hg
I.1.2015.07.H	0.5	0.3	9.7	10.3	28.4	50.0	1.5
I.1.2015.09.H	0.1	0.0	0.8	6.9	33.7	57.1	1.8
I.1.2015.11.H	0.1	0.0	0.1	7.6	42.8	47.3	2.5
I.1.2016.06.H	0.3	0.1	2.0	19.8	31.4	47.1	1.3
I.1.2016.11.H	0.4	0.3	0.8	22.4	44.1	37.0	0.9
I.1.2017.11.H	0.4	0.4	1.0	13.3	57.1	33.6	0.8
II.1.2015.07.H	3.0	0.0	0.2	31.5	25.0	40.3	0.3
II.1.2015.09.H	3.1	0.0	0.1	31.0	29.1	36.7	0.2
II.1.2015.11.H	0.9	0.1	0.3	40.0	24.3	35.8	0.3
II.1.2016.06.H	1.3	0.0	0.1	22.3	50.8	26.1	0.3
II.1.2016.11.H	0.7	0.2	0.6	37.3	41.1	23.8	0.3
II.1.2017.11.H	1.0	0.3	0.7	36.1	37.7	29.7	0.4

[a] The naming convention of the sample ID is based on the following rules: soil type (I or II), greenhouse No. (1 or 2), the sample collection date, and the presence or absence of Hg (H or C).
[b] Percentage of each fraction when total eluted Hg was 100 % (mean, n = 3).

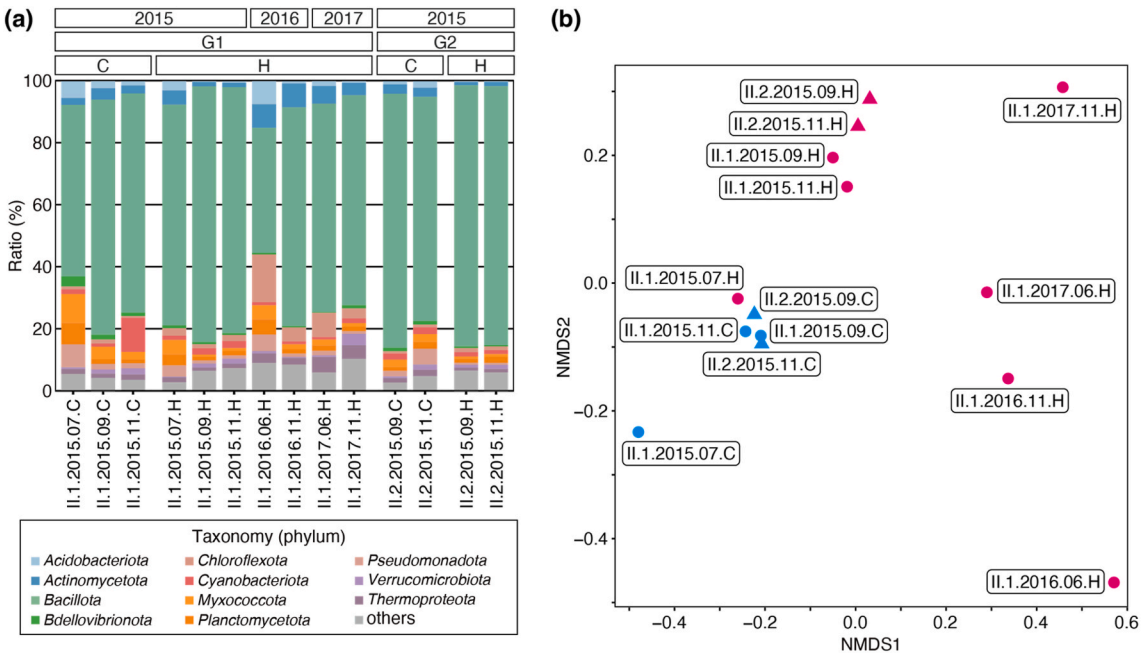


Fig. 3. Influence of Hg addition on microbial compositions in soils analyzed based on the 16S rRNA gene. (a) Microbial community compositions represented by the phylum level classification. DNA extracted from 14 Soil II samples were used for the 16S rRNA gene amplicon sequencing analysis. Taxa with maximum proportion less than 3 % were included in "others". The upper labels indicate the following: The year of sampling, the location of the experiment (G1 or G2), and the presence or absence of Hg (H or C). (b) NMDS ordination plot of the microbial community compositions as determined by amplicon analysis. The NMDS ordination of the 14 soil samples were calculated based on Bray-Curtis dissimilarities and the stress value was 0.019.

marked distinction. Furthermore, the plot location shift was continuously observed in 2016 and 2017. These results suggest that the presence of Hg alters the microbial composition and that this composition changes over time.

Nine out of 14 soil samples underwent metagenome analysis. The occurrence frequencies of the representative housekeeping genes, *recA/*

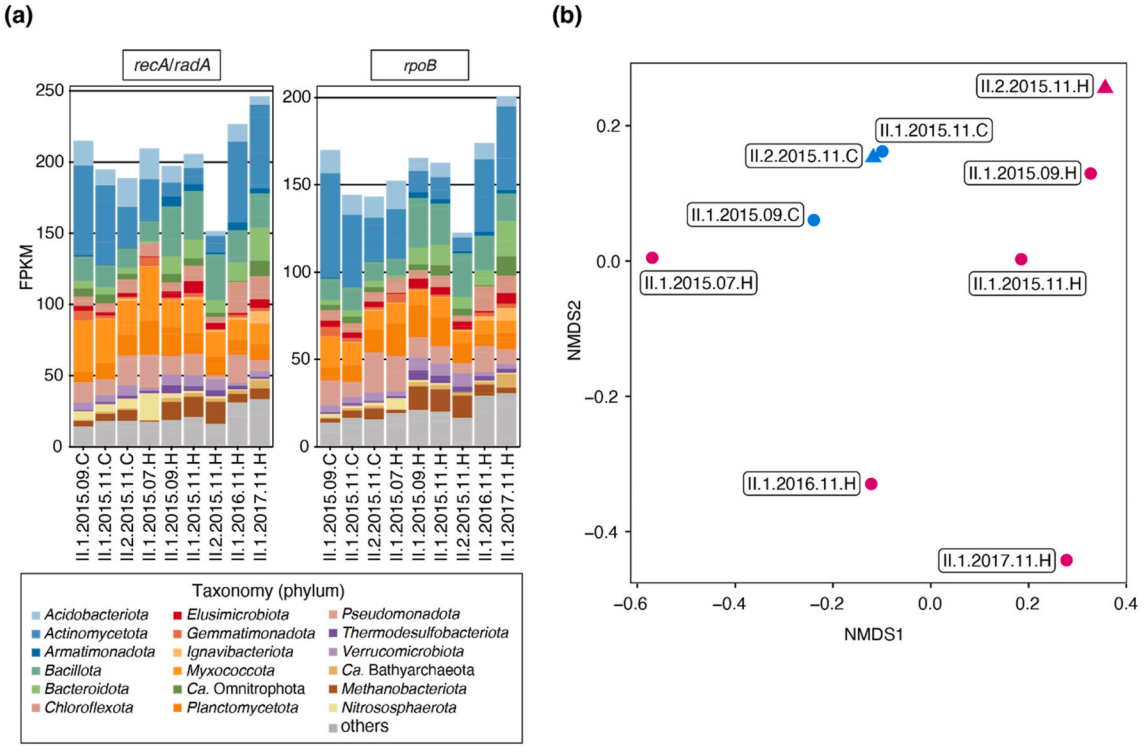


Fig. 4. Influence of Hg^{2+} addition on microbial compositions in soils analyzed based on the housekeeping single copy genes. (a) FPKM for *recA/radA* and *rpoB* genes in the metagenomic analysis were shown along with color coding of the phylum level taxa. Taxa with FPKM less than 3 % of total in each sample were included in "others". (b) NMDS ordination of the microbial community composition based on the FPKM data of the *rpoB* gene. The NMDS analysis was calculated based on Bray-Curtis dissimilarities and the stress value was 0.014. The shape and color of the dots correspond to those indicated in Fig. 3.

radA (bacterial/archaeal DNA recombination protein) and *rpoB* (beta subunit of RNA polymerase), are expressed as fragments per kilobase gene per million sequencing reads (FPKM). Because these genes have only one copy per genome, estimating their abundance ratios in the community is more accurate than analyzing the 16S rRNA gene, which has multiple copies per genome. This analysis showed that the microbial communities were composed of several dominant phyla, including *Actinomycetota, Bacillota, Myxococcota, Planctomycetota, Pseudomonadota, Acidobacteriota,* and *Methanobacteriota,* with relative FPKM values exceeding 10 % (Fig. 4a). This result differed from that of the 16S rRNA amplicon analysis, which indicated exclusive dominance of *Bacillota.* NMDS ordination plots based on the metagenomic *rpoB* gene (Fig. 4b) produced comparable results to those based on the 16S rRNA amplicon. Samples with and without the addition of Hg^{2+} showed different microbial communities. The Hg-containing samples differed from year to year.

A comparison of samples flooded for more than 3 months in the first year (samples of September and November 2015) revealed that several phyla were affected by Hg^{2+} (Fig. S3). Hg^{2+} addition significantly reduced *Actinomycetota, Acidobacteriota,* and the archaeal phylum *Nitrososphaerota*, but *Armatimonadota*, *Bacillota, Methanobacteriota*, and *Thermodesulfobacteriota* increased.

3.5. Functional genes for Hg metabolisms in metagenomic analysis

The frequencies of occurrence of *hgcAB*, *merA*, and *merB* as functional genes for microbial Hg metabolism were investigated using metagenomic data (Fig. 5a). The frequency of occurrence of paired genes *hgcAB* was low in sample II. 1.2015.07. H at the start of the first month of waterlogging and notably increased in samples II. 1.2015.09. H and 11. H, where a suitably anaerobic environment could be deemed to have evolved after 3 months. In addition, the frequency increased in the Hg^{2+}-added samples compared to that in the control samples. The microbial phyla *Bacillota, Thermodesulfobacteriota, Bacteroidota, and*

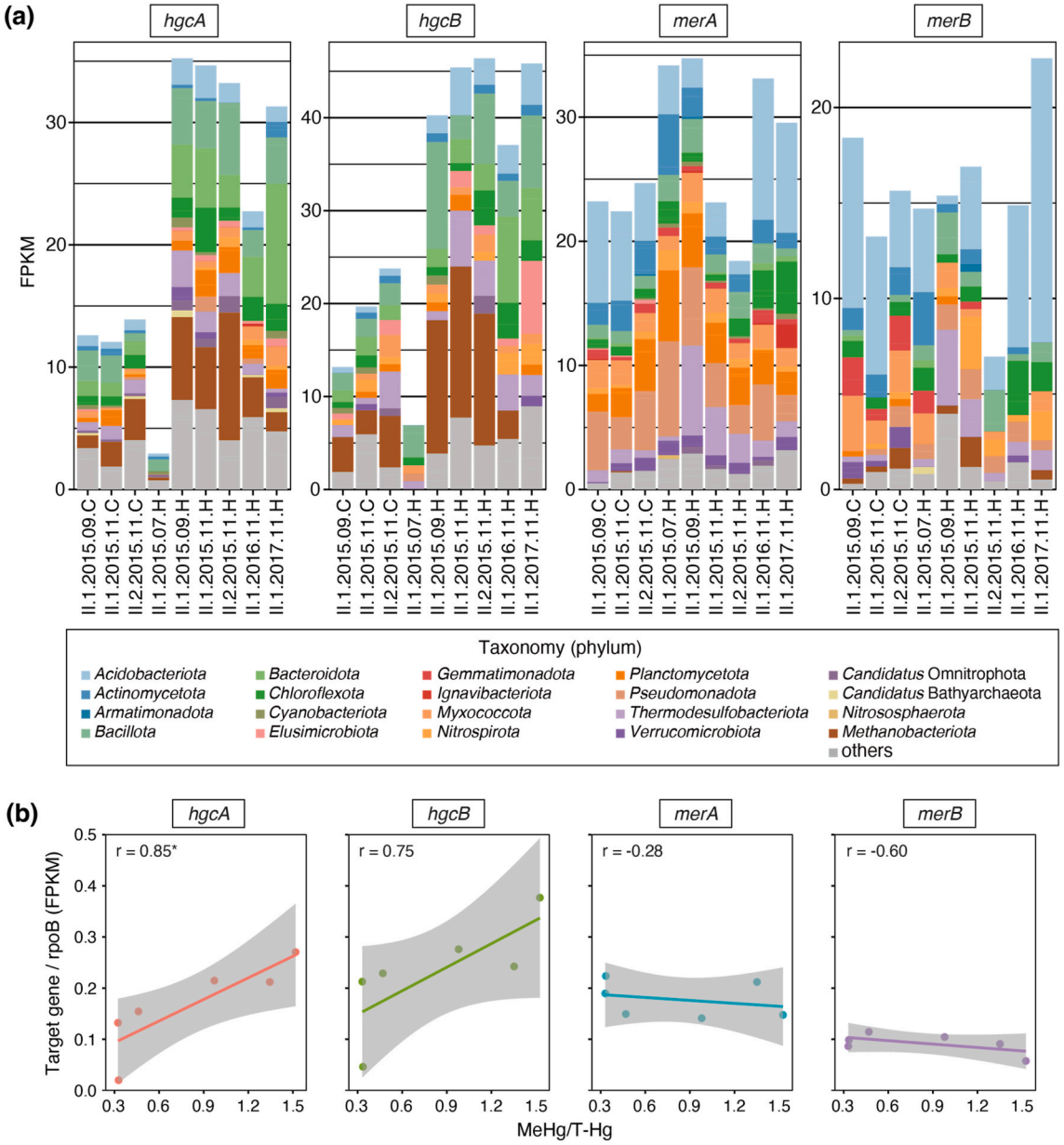


Fig. 5. Metagenomic analyses of the Hg metabolisms functional genes. (a) Occurrence frequency of the Hg metabolisms functional genes, *hgcAB*, *merA* and *merB*. (b) Correlation between occurrence frequency of the target gene and MeHg/T-Hg ratio. Correlation coefficient is expressed as r, and asterisk indicates the P values less than 0.05.

Methanobacteriota contributed to the increase in *hgcAB* genes in Hg-supplemented samples. This is consistent with previous reports that *hgcAB* genes are highly expressed in anaerobic environments, particularly in *Thermodesulfobacteriota* (SRBs), *Methanobacteriota* (methanogens), and *Bacillota* (Liu et al., 2023; Podar et al., 2015), and with current findings that the abundance of these phyla increases with Hg^{2+} addition (Fig. S3). A positive correlation was found between the frequency of occurrence of *hgcAB* genes in the metagenome and the MeHg/T-Hg ratio in the soil (Fig. 5b). The *hgcAB* genes encode corrinoid protein and ferredoxin, which catalyze the methylation of Hg^{2+} to MeHg (Parks et al., 2013). The positive correlation between the abundance of the *hgcAB* genes and the MeHg/T-Hg ratio in the environment is plausible, as previously reported for other environments with varying T-Hg concentrations (Tada et al., 2020). In contrast, we found no clear response in the frequency of the *merA* and *merB* genes to Hg^{2+} addition, nor any correlation between the frequency of these genes and the MeHg/T-Hg ratio (Fig. 5b). The *merA* gene encodes mercuric reductase, which reduces Hg^{2+} to Hg^0, whereas the *merB* gene encodes organomercurial lyase, which catalyzes the demethylation of MeHg to methane and Hg^{2+} (Barkay et al., 2003). This could be because these reactions typically occur in an oxidative environment (Barkay et al., 2003).

We examined changes in the occurrence frequencies of Hg metabolism genes in Hg-added soil samples over three years (II.1.2015.11. H, II.1.2016.11. H, and II.1.2017.11. H), and there was no evident year-to-year difference in the *hgcAB*, *merA,* or *merB* genes (Fig. 5a). By November the peak MeHg production had passed (Fig. 2b), making it difficult to compare year to year.

4. Conclusion

This study tracked rice Hg, soil Hg dynamics, and microbial communities over three years after a single Hg^{2+} addition to two soils. Results from sequential extraction–based speciation of soil Hg suggest that, in addition to soil MeHg concentrations, the mobility of Hg in soil strongly influences Hg accumulation in rice. Moreover, although sequential extraction captured time-dependent shifts in Hg speciation, it did not account for MeHg formation. These findings underscore the limitations of relying solely on sequential extraction to assess Hg present in soils. Hg^{2+} addition persistently altered microbial community composition across years. Metagenomics showed enrichment of *Thermodesulfobacteriota, Methanobacteriota*, and *Bacillota* under Hg, and the abundance of *hgcAB* positively correlated with the soil MeHg/T-Hg ratio, whereas *merA/merB* showed no relationship with MeHg/T-Hg ratio. These results indicate that new Hg contamination in soils alters not only Hg speciation but also the microbial community composition over time, thereby changing the extent of MeHg production and ultimately influencing the level of Hg contamination in rice grains.

Future work will separate plant growth effects, apply multiple flooded–non-flooded cycles, and investigate the relationship between Hg speciation and microbial processes under field-mimicked redox dynamics.

Glossary:

ASGM - artisanal and small-scale gold mining
ASV - amplicon sequence variations
CRM – certified reference material
DOC - dissolved organic carbon
FPKM - fragments per kilobase gene per million sequencing reads
MeHg - monomethylmercury
NMDS – non-metric multidimensional scaling
SRB – sulfate-reducing bacteria
T-C - Total carbon
T-Hg - total mercury
T-N - total nitrogen
T-S – total sulfur

CRediT authorship contribution statement

Hitoshi Kodamatani: Writing – original draft, Methodology, Investigation, Funding acquisition, Conceptualization. **Masahiro Yamamoto:** Writing – original draft, Methodology, Investigation. **Yoshihiro Takaki:** Writing – review & editing, Methodology, Investigation. **Shintaro Hamasuna:** Investigation. **Katsuyuki Ichitani:** Writing – review & editing, Resources, Investigation. **Ryo Kanzaki:** Writing – review & editing, Supervision. **Takashi Tomiyasu:** Writing – review & editing, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2025.144767.

Data availability

Data will be made available on request.

References

Ali, J., Tuzen, M., Shaikh, Q., Jatoi, W.B., Feng, X., Sun, G., Saleh, T.A., Bux, W., Feng, X., Sun, G., Saleh, T.A., 2024. A review of sequential extraction methods for fractionation analysis of toxic metals in solid environmental matrices. TrAC, Trends Anal. Chem. 173, 117639. https://doi.org/10.1016/j.trac.2024.117639.

Barkay, T., Miller, S.M., Summers, A.O., 2003. Bacterial mercury resistance from atoms to ecosystems. FEMS Microbiol. Rev. 27, 355–384. https://doi.org/10.1016/S0168-6445(03)00046-9.

Chen, J., Hu, G., Liu, J., Poulain, A.J., Pu, Q., Huang, R., Meng, B., Feng, X., 2023. The divergent effects of nitrate and ammonium application on mercury methylation , demethylation , and reduction in flooded paddy slurries. J. Hazard. Mater. 460, 132457. https://doi.org/10.1016/j.jhazmat.2023.132457.

Esdaile, L.J., Chalker, J.M., 2018. The Mercury problem in artisanal and small-scale gold mining. Chem. Eur J. 24, 6905–6916. https://doi.org/10.1002/chem.201704840.

Feng, X., Qiu, G., 2008. Mercury pollution in Guizhou, Southwestern China - an overview. Sci. Total Environ. 400, 227–237. https://doi.org/10.1016/j.scitotenv.2008.05.040.

Feng, X., Wang, J., Rinklebe, J., 2024. Methylmercury Accumulation in Rice. CRC Press, Boca Raton. https://doi.org/10.1201/9781003404941.

Fernández-Martínez, R., Loredo, J., Ordóñez, A., Rucandio, I., 2014. Mercury availability by operationally defined fractionation in granulometric distributions of soils and mine wastes from an abandoned cinnabar mine. Environ. Sci. Process. Impacts 16, 1069–1075. https://doi.org/10.1039/C3EM00710C.

Foysal, M.J., Salgar-Chaparro, S.J., 2024. Improving the efficiency of DNA extraction from iron incrustations and oilfield-produced water. Sci. Rep. 14, 1–10. https://doi.org/10.1038/s41598-024-53134-9.

Graham, A.M., Aiken, G.R., Gilmour, C.C., 2012. Dissolved organic matter enhances microbial mercury methylation under sulfidic conditions. Environ. Sci. Technol. 46, 2715–2723. https://doi.org/10.1021/es203658f.

Huang, J.H., Shetaya, W.H., Osterwalder, S., 2020. Determination of (Bio)-available mercury in soils: a review. Environ. Pollut. 263, 114323. https://doi.org/10.1016/j.envpol.2020.114323.

Kanjo, Y., Mouri, M., Kase, T., 2008. Application of modified BCR sequential extraction procedure for evaluation of heavy metal leaching in contaminated soil. J. Japan Soc. Civ. Eng. Ser. G 64, 304–313. https://doi.org/10.2208/jscejg.64.304.

Kodamatani, H., Balogh, S.J., Nollet, Y.H., Matsuyama, A., Fajon, V., Horvat, M., Tomiyasu, T., 2017a. An inter-laboratory comparison of different analytical methods for the determination of monomethylmercury in various soil and sediment samples: a platform for method improvement. Chemosphere 169, 32–39. https://doi.org/10.1016/j.chemosphere.2016.10.129.

Kodamatani, H., Daiba, Y., Morisaki, S., Ichitani, K., Kanzaki, R., Tomiyasu, T., 2020. Detailed investigation of methylmercury accumulation in rice grain from Hg2+-spiked non-contaminated paddy field soils. Chemosphere 247, 125827. https://doi.org/10.1016/j.chemosphere.2020.125827.

Kodamatani, H., Maeda, C., Balogh, S.J., Nollet, Y.H., Kanzaki, R., Tomiyasu, T., 2017b. The influence of sample drying and storage conditions on methylmercury determination in soils and sediments. Chemosphere 173, 380–386. https://doi.org/10.1016/j.chemosphere.2017.01.053.

Kodamatani, H., Matsuyama, A., Saito, K., Kono, Y., Kanzaki, R., Tomiyasu, T., 2012. Sensitive determination method for mercury ion, methyl-, ethyl-, and phenylmercury in water and biological samples using high-performance liquid chromatography with chemiluminescence detection. Anal. Sci. 28. https://doi.org/10.2116/analsci.28.959.

Kodamatani, H., Tomiyasu, T., 2013. Selective determination method for measurement of methylmercury and ethylmercury in soil/sediment samples using high-performance liquid chromatography-chemiluminescence detection coupled with simple extraction technique. J. Chromatogr. A 1288. https://doi.org/10.1016/j.chroma.2013.02.004.

Li, Y., Dai, S.S., Zhao, J., Hu, Z.C., Liu, Q., Feng, J., Huang, Q., Gao, Y., Liu, Y.R., 2023. Amendments of nitrogen and sulfur mitigate carbon-promoting effect on microbial mercury methylation in paddy soils. J. Hazard. Mater. 448, 130983. https://doi.org/10.1016/j.jhazmat.2023.130983.

Li, Y., Lu, C., Zhu, N., Chao, J., Hu, W., Zhang, Z., Wang, Y., Liang, L., Chen, J., Xu, D., Gao, Y., Zhao, J., 2022. Mobilization and methylation of mercury with sulfur addition in paddy soil: implications for integrated water-sulfur management in controlling Hg accumulation in rice. J. Hazard. Mater. 430, 128447. https://doi.org/10.1016/j.jhazmat.2022.128447.

Li, Y., Zhao, J., Zhong, H., Wang, Y., Li, H., Li, Y.-F., Liem-Nguyen, V., Jiang, T., Zhang, Z., Gao, Y., Chai, Z., 2019. Understanding enhanced microbial MeHg production in mining-contaminated paddy soils under sulfate amendment: changes in Hg mobility or microbial methylators? Environ. Sci. Technol. acs.est.8b03511. https://doi.org/10.1021/acs.est.8b03511.

Li, Y.F., Zhao, J., Guo, J., Liu, M., Xu, Q., Li, H., Li, Y.F., Zheng, L., Zhang, Z., Gao, Y., 2017. Influence of sulfur on the accumulation of mercury in rice plant (Oryza sativa L.) growing in mercury contaminated soils. Chemosphere 182, 293–300. https://doi.org/10.1016/j.chemosphere.2017.04.129.

Liu, J., Chen, J., Poulain, A.J., Pu, Q., Hao, Z., Meng, B., Feng, X., 2023. Mercury and sulfur redox cycling affect methylmercury levels in rice paddy soils across a contamination gradient. Environ. Sci. Technol. 57, 8149–8160. https://doi.org/10.1021/acs.est.3c02676.

Liu, J., Lu, B., Poulain, A.J., Zhang, R., Zhang, T., Feng, X., Meng, B., 2022a. The underappreciated role of natural organic matter bond Hg(II) and nanoparticulate HgS as substrates for methylation in paddy soils across a Hg concentration gradient. Environ. Pollut. 292, 118321. https://doi.org/10.1016/j.envpol.2021.118321.

Liu, J., Wang, J., Ning, Y., Yang, S., Wang, P., Shaheen, S.M., Feng, X., Rinklebe, J., 2019a. Methylmercury production in a paddy soil and its uptake by rice plants as affected by different geochemical mercury pools. Environ. Int. 129, 461–469. https://doi.org/10.1016/j.envint.2019.04.068.

Liu, J., Zhao, L., Kong, K., Abdelhafiz, M.A., Tian, S., 2022b. Uncovering geochemical fractionation of the newly deposited Hg in paddy soil using a stable isotope tracer. J. Hazard. Mater. 433, 128752. https://doi.org/10.1016/j.jhazmat.2022.128752.

Liu, M., Zhang, Q., Cheng, M., He, Y., Chen, L., Zhang, H., Cao, H., Shen, H., Zhang, W., Tao, S., Wang, X., 2019b. Rice life cycle-based global mercury biotransport and human methylmercury exposure. Nat. Commun. 10. https://doi.org/10.1038/s41467-019-13221-2.

Ndu, U., Christensen, G.A., Rivera, N.A., Gionfriddo, C.M., Deshusses, M.A., Elias, D.A., Hsu-Kim, H., 2018. Quantification of mercury bioavailability for methylation using diffusive gradient in thin-film samplers. Environ. Sci. Technol. 52, 8521–8529. https://doi.org/10.1021/acs.est.8b00647.

Novirsa, R., Dinh, Q.P., Jeong, H., Addai-Arhin, S., Nugraha, W.C., Hirota, N., Wispriyono, B., Ishibashi, Y., Arizono, K., 2020. The dietary intake of mercury from rice and human health risk in artisanal small-scale gold mining area, Indonesia. Fundam. Toxicol. Sci. 7, 215–225. https://doi.org/10.2131/fts.7.215.

Novirsa, R., Quang, P., Jeong, H., Fukushima, S., Ishibashi, Y., Wispriyono, B., Arizono, K., 2019. The evaluation of mercury contamination in upland rice paddy field around artisanal small-scale gold mining area, Lebaksitu, Indonesia. J. Environ. Saf. 10, 119–125. https://doi.org/10.11162/daikankyo.E19RP0103.

O'Connor, D., Hou, D., Ok, Y.S., Mulder, J., Duan, L., Wu, Q., Wang, S., Tack, F.M.G., Rinklebe, J., 2019. Mercury speciation, transformation, and transportation in soils, atmospheric flux, and implications for risk management: a critical review. Environ. Int. 126, 747–761. https://doi.org/10.1016/j.envint.2019.03.019.

Obrist, D., Kirk, J.L., Zhang, L., Sunderland, E.M., Jiskra, M., Selin, N.E., 2018. A review of global environmental mercury processes in response to human and natural perturbations: changes of emissions, climate, and land use. Ambio 47, 116–140. https://doi.org/10.1007/s13280-017-1004-9.

Parks, J.M., Johs, A., Podar, M., Bridou, R., Hurt, R.A., Smith, S.D., Tomanicek, S.J., Qian, Y., Brown, S.D., Brandt, C.C., Palumbo, A.V., Smith, J.C., Wall, J.D., Elias, D. A., Liang, L., 2013. The genetic basis for bacterial mercury methylation. Science 339, 1332–1335. https://doi.org/10.1126/science.1230667.

Pelcová, P., Ridošková, A., Hrachovinová, J., Grmela, J., 2020. Fractionation analysis of mercury in soils: a comparison of three techniques for bioavailable mercury fraction determination. Environ. Toxicol. Chem. 39, 1670–1677. https://doi.org/10.1002/etc.4797.

Podar, M., Gilmour, C.C., Brandt, C.C., Soren, A., Brown, S.D., Crable, B.R., Palumbo, A. V., Somenahally, A.C., Elias, D.A., 2015. Global prevalence and distribution of genes and microorganisms involved in mercury methylation. Sci. Adv. 1, 1–12. https://doi.org/10.1126/sciadv.1500675.

Pu, Q., Zhang, K., Liu, J., Zhang, Q., Abdelhafiz, M.A., Meng, B., Feng, X., 2024. Key active mercury methylating microorganisms and their synergistic effects on methylmercury production in paddy soils. J. Hazard. Mater. 480, 136481. https://doi.org/10.1016/j.jhazmat.2024.136481.

Reis, A.T., Davidson, C.M., Vale, C., Pereira, E., 2016. Overview and challenges of mercury fractionation and speciation in soils. TrAC, Trends Anal. Chem. 82, 109–117. https://doi.org/10.1016/j.trac.2016.05.008.

Roose-Amsaleg, C.L., Garnier-Sillam, E., Harry, M., 2001. Extraction and purification of microbial DNA from soil and sediment samples. Appl. Soil Ecol. 18, 47–60. https://doi.org/10.1016/S0929-1393(01)00149-4.

Soe, P.S., Kyaw, W.T., Arizono, K., Ishibashi, Y., Agusa, T., 2022. Mercury pollution from artisanal and small-scale gold mining in Myanmar and other Southeast Asian countries. Int. J. Environ. Res. Publ. Health 19. https://doi.org/10.3390/ijerph19106290.

Tada, Y., Marumoto, K., Takeuchi, A., 2020. Nitrospina-like bacteria are potential mercury methylators in the mesopelagic zone in the East China Sea. Front. Microbiol. 11. https://doi.org/10.3389/fmicb.2020.01369.

Tomiyasu, T., Baransano, C., Hamada, Y.K., Kodamatani, H., Kanzaki, R., Hidayati, N., Rahajoe, J.S., 2020. Distribution of total and organic mercury in soils around an artisanal and small-scale gold mining area in West Java, Indonesia. SN Appl. Sci. 2, 1–11. https://doi.org/10.1007/s42452-020-3008-5.

Tomiyasu, T., Hamada, Y.K., Kodamatani, H., Hidayati, N., Rahajoe, J.S., 2019. Transport of mercury species by river from artisanal and small-scale gold mining in West Java, Indonesia. Environ. Sci. Pollut. Res. 26, 25262–25274. https://doi.org/10.1007/s11356-019-05718-6.

Tomiyasu, T., Kono, Y., Kodamatani, H., Hidayati, N., Rahajoe, J.S., 2013. The distribution of mercury around the small-scale gold mining area along the Cikaniki river, Bogor, Indonesia. Environ. Res. 125. https://doi.org/10.1016/j.envres.2013.03.015.

Wadle, A., Neal-Walthall, N., Ndu, U., Hsu-Kim, H., 2024. Distribution and homogenization of multiple mercury species inputs to freshwater wetland mesocosms. Environ. Sci. Technol. 58, 1709–1720. https://doi.org/10.1021/acs.est.3c07169.

Wang, Y., Dang, F., Zheng, X., Zhong, H., 2019. Biochar amendment to further reduce methylmercury accumulation in rice grown in selenium-amended paddy soil. J. Hazard. Mater. 365, 590–596. https://doi.org/10.1016/j.jhazmat.2018.11.052.

Xia, J., Wang, J., Zhang, L., Wang, X., Yuan, W., Anderson, C.W.N., Chen, C., Peng, T., Feng, X., 2021. Significant mercury efflux from a Karst region in Southwest China - results from mass balance studies in two catchments. Sci. Total Environ. 769, 144892. https://doi.org/10.1016/j.scitotenv.2020.144892.

Yang, Q., Wang, Y., Zhong, H., 2021. Remediation of mercury-contaminated soils and sediments using biochar: a critical review. Biochar 3, 23–35. https://doi.org/10.1007/s42773-021-00087-1.

Zheng, Z., Hu, J., He, T., Liu, C., Zhou, X., Yin, D., 2024. Suppression of mercury methylation in soil and methylmercury accumulation in rice by dissolved organic matter derived from sulfur-rich rape straw. Environ. Pollut. 346, 123657. https://doi.org/10.1016/j.envpol.2024.123657.

Zhong, H., Tang, W., Li, Z., Sonne, C., Lam, S.S., Zhang, X., Kwon, S.Y., Rinklebe, J., Nunes, L.M., Yu, R.-Q., Gu, B., Hintelmann, H., Tsui, M.T.-K., Zhao, J., Zhou, X.-Q., Wu, M., Liu, B., Hao, Y., Chen, L., Zhang, B., Tan, W., Zhang, X.-X., Ren, H., Liu, Y.-R., 2024. Soil Geobacteraceae are the key predictors of neurotoxic methylmercury bioaccumulation in rice. Nat. Food 5, 301–311. https://doi.org/10.1038/s43016-024-00954-7.

Zvěřina, O., Coufalík, P., Komárek, J., Gadas, P., Sysalová, J., 2014. Mercury associated with size-fractionated urban particulate matter: three years of sampling in Prague, Czech Republic. Chem. Pap. 68, 197–202. https://doi.org/10.2478/s11696-013-0436-3.